

Department of Civil Engineering
(MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY ALLAHABAD)
B. Tech VI Semester
CEN16103 (Design of Hydraulic Structures)
Tutorial Sheet (s): 5-6-7-8
Due Date: On or Before April 23, 2026

Instructions:

- (i) **Assume any data if required.**
- (ii) **Draw annotations wherever required.**
- (iii) **All questions are mandatory.**
- (iv) **Other Numerical problems solved on SK Garg book may be submitted in excel sheet in addition to given problems for additional advantage.**

Tutorial-5

(1) A Sarda fall for a channel is carrying $45 \text{ m}^3/\text{s}$ of water at a depth of flow equal to 1.8 m. The bed width of the channel is 28 m. The following data is associated with the fall:

	Upstream	Downstream
Full Supply Level (m) =	118.30	116.80
Drop (m) =		1.50
Bed level (m) =	116.50	115.00

Design (a) the dimensions of the crest and (b) cistern downstream of the fall.

(2) Design a guide bank required for a bridge on a river having following particulars:

Design flood discharge= 50000 cumecs ; Silt factor= 1.1 ; Bed level of river= 130.0 m ; High flood level= 140.0 m

Also, find the volume of stone required per m length of the guide bank

(3) The head regulator of a canal has 3 openings each 3 m wide. The water is flowing between the upper and lower gates. The vertical opening of the gate is 1.0 m. The head on the regulator is 0.45 m (Afflux). If the upstream water level rises by 0.20 m, find how much the upper gates must be lowered to maintain the canal discharge unaltered.

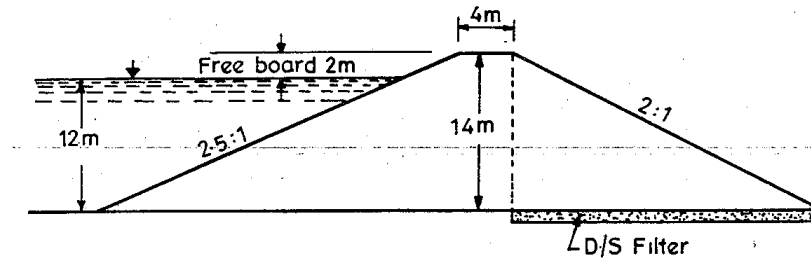
Tutorial-6

(4) A concrete gravity dam has top width 8 m and height 40 m above the base. Freeboard is 4 m and the width at the base is 36 m. Upstream face of the dam is vertical, but the downstream face is sloping uniformly up to the base from a height 10 m below the top of the dam. Determine:

- (i) Factor of safety against overturning.
- (ii) Factor of safety against sliding ($\mu=0.75$).
- (iii) Maximum principal stress at the toe.

Consider only the forces due to self-weight, uplift (full) and hydrostatic thrust. Take unit weight of concrete as 21.582 kN/m^3 (2.2 t/m^3)

(5) A section of a homogeneous earth dam is shown in the Fig. shown below. Calculate the seepage discharge per metre length, through the body of the dam. The coefficient of permeability of the dam material may be taken as 8×10^{-5} m/sec.



(6) A masonry dam 80 m high is trapezoidal in section with top width 2 m. The face exposed to water has a slope of 1 horizontal to 10 vertical and the downstream face has a slope of 0.75 horizontal to 1 vertical. Test the stability of the dam at its base. Also find the principal stresses at the toe and heel of the dam and shear friction factor:

(i) Assume tail water of depth 5 m.

(ii) Assume a line of drain at 6 m from the heel and a free board of 5 m.

(iii) Take $\mu=0.65$; specific gravity of masonry as 2.40 and allowable shear stress as 490.5 kN/m^2 (50 t/m^2).

(iv) Horizontal acceleration due to earthquake may be taken as $0.1g$.

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7. Compute the discharge over an ogee spillway with coefficient of discharge $c = 2.3$ at a head of 3.8 m. The effective length of the spillway is 110 m. Neglect the velocity of approach

8. Design a suitable section for the overflow portion of a concrete gravity dam having the downstream face sloping at $0.7H : 1V$. The design discharge for the spillway is 8000 cumecs. The height of the spillway crest is kept at RL 204.0 m. The average river bed level at the site is 100.0 m. The spillway consists of 6 spans and $L = 60\text{m}$. Thickness of each pier may be taken to be 2.5 m. Neglect velocity of approach. Assume the coefficient of the spillway $C = 2.2$. Assume the pier contraction coefficient and the abutment contraction coefficient to be $K_p=0.01$, $K_a=0.10$.

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9. A run-off river plant is installed on a river having a minimum flow of 15 cumecs. If the plant is used as a peak load plant operating only for 6 hours daily, compute the firm capacity of the plant :

(a) without pondage ;

(b) with pondage but allowing 8% water-to be lost in evaporation and other losses. Head at the plant is 16 m, and the plant efficiency may be assumed as 80%.

10. During a low water week, a river stream gets average daily flow of 35 cumecs with daily fluctuations requiring a pondage capacity of 16% of the daily flow. A hydroelectric plant is to be located on this river which will operate for 5 days a week , 24 hours a day, but will supply power at varying rates, such that daily load factor is of 60%, corresponding to which the pondage required is 0.2 time the mean flow to the turbines. On Saturdays and Sundays, all the flow is ponded for use on rest of the days. If the effective head on the turbine, when the pond is full is to be 26 m, and the maximum allowable fluctuations on the pond level is 1.0 m, determine:

(a) The surface area of the pond to satisfy all the operating conditions;

(b) The weekly output at the switch boards in kWh.

Assume turbine efficiency=82%; Generator efficiency=90%.